Central Topic

6

Main Topic 4

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Economy Agriculture

Macro Data of Agriculture

% of GDP 18%

Growth rate over last 5 years 4.2%

% of total exports

11.7%

$53 Bn

% of workforce 45%

Female nearly 47% in agriculture were force
but only 12.8% are land owner agriculture

avg farm size

1970-71 2.28 hectare

2015-16 1.08 hectare

USA 150 Ha

Parlance

water intensive water guzzling crop

dropiing water levels conversion of indo-gangetic plain into dark

zones

'bait to plate' tracaibility using IT technology in
aquaculture

human back to machine back via shift from
bag packing to containerisation

fork to farm

farm to gate

farm to fork

sustainable agriculture

reduced checmical residue in food and
biomagnification

water intensive to high value horticulture crops

This necessitates shared economy models like
"Uberisation" of farm services

Evergreen revolution coined by MS
Swaminathan

to address shortcoming of Green Revolution

producing more with less land, less water, less
pesticides

to achieve sustainiable agriculture

NCF National commssion on Farmers swaminathan

DFI dalwai committee

high level committe HLC shanta kumar

Major crops-cropping patterns in various parts
of the country, - different types of irrigation and
irrigation systems storage, transport and
marketing of agricultural produce and issues
and related constraints; e-technology in the aid
of farmers.

Major crops-cropping patterns in various parts
of the country,

Cropping Pattern

refers to sequence, arrangement and
proportion of different crops over a period of
time

Definitions

Crop Rotation changing crops every year on a particular area

Inter cropping growing two or more crops together

relay cropping sowing another crop before one is harvested

sequence cropping sowing another crop after one is harvested

Monocropping one crop ona field

AGROECOLOGY

FAO defines it as a science, a set of practices,
and a social movement aiming at sustainable
food systems.

"Farming with nature, not against it."

Agroecology is an ecological approach to
agriculture that applies principles of ecology,
biodiversity, sustainability and social equity to
the design and management of agri-food
systems

REGENERATIVE FARMING

Regenerative farming is an agricultural
approach aimed at restoring soil health,
enhancing biodiversity, improving water cycles,
and sequestering carbon, while maintaining
farm profitability.

factors responsible for cropping pattern

MSP

Market demands

subisdies electricity, fertiliser

irrigation support

seed availability

Rice wheat mono culture issue

depleting water levels in northern plains

rising soil acidity and decline in organic matter

stubble burning deteroriates envt health and
ecosystem

decline in crop diversity thus food habits shift
away from

increase in vulnearbility to climate change and
weather socks

Recent changes in cropping patter

Shift from cereals to high value crops like
Horticulture fruits vegetables spices driven by
high profitability and consumer demand

Decline in course grains like jowar and bajra
due to low msp, poor procurement and
changing food habits

Pulses are witnessing a revival due to
improved msp and government promotion

Expansion of commercial crops like sugarcane
,cotton and oil seeds

Farmings

Zero Budget Natural Farming

The term zero budget natural farming was
coined by Subhash Palekar, it means raising
crops without using any fertilizers and
pesticides or any other external materials

Method

Bijamrit - the microbial coating of seeds with
formulations of cow urines and cow dungs.

Jivamrit is the enhancement of soil microbes
using inoculum of cow dung, cow urine and
jaggery

Mulching is the covering of soil with crops and
crop residues which creates humus and
encourages the growth of friendly
microorganisms.

reduction in soil erosion and imprve soil health

promotes carbon capture

reduce labour (zero tillage)

Waaphasa is the building up of soil humus to
increase soil aeration.

It requires cow dung and cow urine from Indian
breeds cow only

Multi cropping is encouraged

Schemes

Paramparagat Krishi Vikas Yojana (PKVY)

Supports adoption of natural farming practices
through cluster formation, training, and
certification of group organic production.

National Mission on Natural Farming (NMNF)

Nodal programme to scale up ZBNF across
states with institutional support, knowledge
dissemination, and input formulation centres.

Under National Mission for Sustainable

Agriculture (NMSA), ZBNF practices are
promoted for soil health, water efficiency, and
climate resilience.

Organic Farming

uses approved natural inputs (compost,
manure, biofertilizers) and follows certified
standards

Benefits

improved market access to premium health
consious consumers

improved soil health and ecosystem restoration

reduced chemical residue in food, enhance
consumer health and food safety

Lower environmental footprint, minimizing soil,
water, and air pollution.

Promotion of biodiversity, including beneficial
insects and soil organisms.

Alignment with climate goals, as organic soils
act as carbon sinks and reduce synthetic
fertilizer emissions.

Schemes

Paramparagat Krishi Vikas Yojana (PKVY)

Supports adoption of natural farming practices
through cluster formation, training, and
certification of group organic production.

National Programme for Organic Production
(NPOР)

Certification framework (NOP/PGS-India) to
ensure market recognition and export
standards for organic produce.

Mission Organic Value Chain Development for
North East Region (MOVCDNER)

Value chain creation for organic produce,
including collection centres, processing units,
and market access (now expanding beyond
North East).

organic vs natural farming

uses approved natural inputs (compost,
manure, biofertilizers) and follows certified
standards

natural farming (like ZBNF) is a more
minimalist, zero-input approach, avoiding even
organic additives, relying solely on nature's
processes, cow dung/urine, and on-farm
biomass for soil health, focusing on no tilling,
mulching, and self-sufficiency over external
inputs or certification

Organic farming is regulated and can involve
purchased inputs

No purchased inputs are allowed

Uses crop rotation, composting, and may still
involve tilling and weeding

no tilling, ploughing,

Can be capital-intensive due to input costs. is minimal in production cost

Requires strict third-party certification (e.g.,
APEDA in India).

Generally follows principles, not formal
certification.

Integerated Farming

An Integrated Farming System is a sustainable
and holistic approach to agriculture that
combines crop production with allied activities
such as livestock, poultry, fishery, agroforestry,
apiculture, sericulture, etc, to optimize
resource useand enhance income.

Precision Farming

Farming in which inputs are provided with
precision and proper quantitiy with the helpm
of sensors, ICT, variable discharge technoligies

GM crops

intro

India's government approved the conditional
environmental release of GM mustard hybrid
DMH-11 in October 2022 for seed production
and testing

According to WHO, genetically modified crops
are crops derived from organisms whose
genetic material (DNA) has been modified in a
way that does not occur naturally,

Benefits of GM crops

Tackling food security challenges

2. Higher yield

3. Climate change resilience

4. Pest resistance

5. Tackle nutrition challenges

Challenges of GM Crops

1. Doubtful yield increase: The increase in yield
by GM crops is still not proven.

2. Terminator seeds, which produce infertile
seeds at harvest thereby increasing the

dependence of farmers on seed companies.

3. Resistance to GM crops developed in pests

4. Potential threat to environment and health in
the long term.

Way forward

1. Closed field trials

2. reforming gesc to be headed by expert from
field of biotech and

3. assessing long term effect on environment
and human health

4. Labeling GM products

5. Geac should be given statutory status to
provide autonomy to its functions

6. Recommended legislation relating to liability
and redress for damage due to living modified
organisms to comply with nagoya protocol on
biosafety

Supreme Court delivered a split verdict in July
2024, calling for a national GM crop policy

Framework

sustainiablitiy

biodiversity promotion thorugh increase in
benefiting insects and microrganisims,

reduction in water, air pollution,

improved soil health,

economic

prodcutivity

, resource optimisation

better exclusive premium consumer market
access

Schemes PM FASAL

In case of failure due to natural calamities,pest
attack etc

Central sponsored scheme

Horti/livestock/aquaculture/food processing

% of GVA

33

30

2014-15 24%

2022-23 30%

6.7

FPI

agri 11.5%

manufacturing 10%

CAGR (Decadal)

2.82% (Production wise)

7.13%

6.75%

5.35% (2015-23)

% of Agri exports

7.5

9.3

15

23%

Employement

-

21 Mn

28 Mn

70L / 12% of formal workforce

Mission

Horti

Mission for Integrated Development of
Horticulture (MIDH)

fruits,vegetables, flowers from prod to post
harvest

storage infra

cold supply chains, storage houses, processing
unitsetc.

cluster formation, fpo, direct marketing

trainign in horticulture

tech and R&D support

National Horticulture Board (NHB) post harvest infra in integerated manner

horticulture CDP

Livestock

NDDB promote farmer owned dairy and cooperatives

Rashtriya Gokul Mission

indigenous bovines and enhance milk
produciton and producivity

National Livestock Mission sustainable dev of livestocks

Animal husbandry infrastructure development
fund amounting to 15000 crores.

Aquaculture

PM Matsya Sampada Yojna Blue revolution

production, quality

sustainable and responsible

Fisheries and Aquaculture Infrastructure
Development Fund (FIDF)

fish inra harbours to cold chains

Concerns

Horti

Climate change, water scarcity, post-harvest
losses

Livestock

Greenhouse gas emissions, resource depletion,
low productivity

Aquacultre

Environmental impact (mangrove loss), water
pollution, disease outbreaks

Crops and produce

Cereal

2023 332 mt

1947 50 mT

Expected rise this year

6.6%

354 mt wheat and rice alone constitute 78% of food grains

Productivity

Wheat 3.4-3.6 Ton

Rice 2.8 Ton

India ~3 tonne / hectare

China ~6.1 tonne/ hectare

FAO

General Schemes PM FASAL

in case of crop failure by pest or natural
calamity

Millets

Demand per capita per month in rural india

1962 4.4Kg

1983 3.3 Kg

2011 0.4Kg

90% decline from pre green reovlution

Export 2nd Largest in world

Productivity 1.9 tonne per hectare

Rise as IYoM capagain 40% growth in sales Amazon

why millets

Climate Resilence <30 cm rain

staple crop for tribals 60% SECC

Benefits Health

Gluten free ; suitable for people with gluten intolearnace

high nutrients

Ragi 10x calcium than wheat or Rice

Bajra Iron

low glycemic index help in controlling blood sugar levels

Anti-oxidants

improve digestive health

high fibre content aids in digestion and
prevents costipation

Heart health

Mg and K help in lowering blood pressure and
reduce eart disease risk

addreses malnutrition

Conclusion Poor Man Grain -> Smart Food

MoAFW

Horticulture

352 mt 2023

2nd largest in world Production

pick 4 vital stats from above + change in area

Why a good sector

High income 5-10x in apple relative to wheat

Labour Intensive planting, harvesting, etc.

Nutritional Security Hidden Hunger

Exports $3.8 Bn

Challenges

Low Processing

2-3% in fruits

Global 60%

MoSPI

Post Harvest Loses 20%

Fragmented Supply Chain

Price Volatility

Credit Flow <5% of total agri credit

Climate Change Vulnearabilty

% of agri gva of R&D <1%

Govt Stance

MIDH (subsumes NHM)

production

infra

market

National Horticulture Board

Financial assistance for cold storage

Nurseris

Operation Greens : TOP -> TOTAL (incudes
fruits and vegetables)

subsidies on transport and storage

Horticulture CDP

Way Forward

CDP Kerala CDP Jackfruit

FPO promotion and contract farming Banana in Jalgaon

Digitisation

Pest forecasting

AI Advisory

Kisan Sarthi App

Export oriented porduction Grapes/Litchis

Cold Storage through solar power

Crop Insurance

Pulses

Protein by weight 20-25%

Largest

Producer 25%

Consumer 27%

Importer 14%

Import dependence 25%

Solution:

Integrated Schemes of Oilseeds, Pulses, Oil
palm and Maize or ISOPOM,

Oil seeds Import Dependence 55%

Cropping Intensity

Area Under Food Grain (Cross Cropped Area)

1980 73%

2022 62%

Coarse cereals Area

1980 22%

2022 9%

Oil Seed

1980 9%

2022 15%

HortiCulture

2001 8.5%

2022 14%

Pulses

are seeing revival due to improved MSP and
govt interventions

Gross cropped area / net sown area

1950 111%

2019-20 151%

change in dietary needs, exports and urban
demand

MoAFW

different types of irrigation and irrigation
systems storage

Irrigation

Intro land under irrigation of Gross Cropped Area

2016 49%

2021 55%

NITI Aayog

Data

Rainfed Agriculture % of Prduction

Millets 90%

Oil Seeds 80%

Livestock/Cotton 60%

%of irrigated area under groundwater irrigation 60%

%of ground water usage for irrigation 90%

% of global fresh water resource in India 4%

rice and sugarcane take 60% of water available for irrigation

ultimate irrigatoiin potential in India 140 Mn hectare

created irrigation potential in India 99 Mn Hectare

Types

Major (surface)

canals

Storage (59Mn Hectare)

large dams

large reservoirs

significant potential is created while utlisation
lags behind

Minor (surface and ground water)

Ground water

60% of total irrigated area

It flourished because public canal systems
failed to deliver reliable water on demand

surface minor irrigation tanks and water bodies

Traditional systems like tanks, khadins, and
nadis are crucial for small/marginal farmers,
particularly in Southern and Western India.

these systems have declined due to siltation
and encroachments

Micro irrigation

Includes Drip and Sprinkler systems aimed at
high water-use efficiency ("Per Drop More
Crop")

Sub-surface drip irrigation is also being tested
to reduce evaporation losses

Issues

high dependence on monsoon

gap between potential created and utilised ~25 Mn Hectare due to

poor maintainence of distribution channel

lack of field connectivity

DFI

inefficient water use

traditional methods such as flood irrigation has
efficiency of only 35-40%

leads to methane emission, soil erosion and
eutrophication

overextraction of groundwater leading to dark
zones where extraction exceeds recharge

High initial Costs of Modern Irrigation
Systems: drip and sprinkler systems

Inter-State Water Disputes: Conflicts over river
water sharing, the Cauvery and Krishna rivers,
hinder coordinated irrigation planning and
managemen

Decline of Traditional Systems: Traditional tank
irrigation has declined due to encroachment on
catchment areas, siltation, and the breakdown
of community management institutions

Inequity: Head-reach farmers in canal
commands often consume excessive water,
leaving tail-end farmers deprived. This leads to
salinity/waterlogging at the head and scarcity
at the tail

Govt stance

PMKSY

Accelerated irrigation benefit Programme

Focuses on completing long-pending major
and medium irrigation projects

per drop mroe crop (efficiency)

Har Khet ko Pani (water for every field) command area developemnt

integerated watershed development

Jal Shakti Abhiyan: A time-bound campaign
focusing on rainwater harvesting, renovation of
water bodies, and watershed development

Solarisation (PM-KUSUM): Promoting solar
pumps to reduce dependence on the grid and
diesel. This also allows farmers to sell surplus
power, theoretically incentivizing them to use
less water

Micro-Irrigation Fund (MIF): A corpus of ₹5,000
crore was created with NABARD to help states
mobilize resources for expanding micro-
irrigation coverage

Recommended reforms

Microirrigation

can save upto [Ashok Dalwai Committee] 40%
of water

Micro-irrigation is a precise watering method
delivering water slowly as drops, streams, or
mist directly to the plant's root zone using
emitters like drippers or sprinklers,
dramatically increasing water efficiency (45-
60% less water than surface irrigation)

Benefits

Water use efficiency and prevents water logging

2. Energy Efficiency

3. Increase in crop productivity

Issues

Under utilization of potential

2. huge primary cost

3. delays in subsidy payment

4. knowledge gap

5. smaller and fragmented land Holdings

Major cause of water inefficiency [NITI Aayog]

Electricity subsidy

Use graded Metered electricity

There is an urgent need to align cropping
patterns with agro-climatic zones.

Water-guzzling crops like rice and sugarcane
should be shifted away from water-scarce
areas (e.g., Punjab, Maharashtra) to water-
surplus regions

Promoting less water-intensive crops like
millets (nutri-cereals), pulses, and oilseeds in
rainfed areas

Mandatory Micro-Irrigation: Micro-irrigation
should be made mandatory in the command
areas of irrigation projects and for water-
intensive crops to improve efficiency

Participatory Irrigation Management (PIM):

Management of irrigation systems should be
transferred to Water Users Associations

(WUAs) and Panchayats.

They should handle water distribution and fee
collection to ensure equity and maintenance

Use of sensors, IoT, and automation for
precise irrigation scheduling based on soil
moisture (e.g., e-IAS, IkshuKedar app)

Remote sensing and GIS for monitoring water
bodies and planning watershed interventions

Restoration of Water Bodies: A massive drive
to desilt and renovate village tanks and ponds
(e.g., Mission Kakatiya model) to act as
storage buffers and recharge groundwater

Watershed Management: Adopting a
comprehensive watershed approach ("Store
Grain and Water everywhere") to conserve soil
and water, particularly in rainfed areas

Gujarat bhungroo is a water management
system that injects and stores access rainfall
water underground

transport and marketing of agricultural
produce and issues and related constraints

Fork to farm apporach

recommened by ashok dalwai committee

undertake cultivation of crops upon prevailing
demand

transport of agricultural produce and issues
and related constraints

status

Approximately 97.4% of perishable fruits and
vegetables are transported by road, while only
1.9% move via the rail network. For general
cargo, roads handle 66% while railways handle
26%

FCI food grain

80% moved via rail

19% via roads

Cold Chain Gap: There is a severe deficit in
cold chain infrastructure required for
perishables

Issues

Rake Shortage: A major constraint is the
shortfall in the supply of railway rakes.

During the audit period (2017-22), there was a
shortfall in movement ranging from 17% to
19% against the planned quantity due to
constraints like non-availability of rakes and
route restrictions

High Costs & Demurrage: Delays in
loading/unloading lead to high demurrage
charges. In a five-year period, demurrage
charges levied by Railways amounted to
₹211.55 crore, of which only a small fraction
was waived

The movement is largely largely done via
conventional wagons rather than containers,
leading to higher transit losses and slower
turnaround times

Inadequate Bulk Handling:

Despite plans to modernize, the system still
heavily relies on manual handling of gunny
bags ("human back" to "machine back"
transition is slow)

Even where silos exist, wheat is often received
in bags rather than in bulk, negating the
efficiency of modern storage

Silo Connectivity: Some modern silos
constructed under Public-Private Partnerships
(PPP) lack necessary railway sidings or
dispatch facilities, rendering them unable to
liquidate stocks efficiently.

Transit Losses: Losses occur due to multiple
handlings, use of iron hooks on bags,

Between 2017 and 2022, transit losses
amounted to 5.76 LMT, valued at approximately
₹1,651 crore

While road transport is essential for last-mile
connectivity, it faces challenges such as the
high cost of interstate movement and the lack
of a unified national market structure

First-Mile Connectivity: The link between the
farm gate and the first point of aggregation
(PRAM/GrAM) is weak.

Small farmers often lack access to efficient
transport to reach wider markets

Geographical Constraints: Movement to
difficult terrains, such as the North-Eastern
states and Jammu & Kashmir, remains
logistically complex and costly

Govt stance

Hub and Spoke Model: The government
promotes a new market architecture linking
Primary Rural Agricultural Markets (GrAMs) to
wholesale markets (APMCs) and export hubs
to streamline logistics

National Policy on Handling, Storage, and
Transportation (2000): The government aims
to modernize the system by encouraging bulk
handling, silos, and private sector participation

PM Gati Shakti: This national master plan aims
to provide multimodal connectivity
infrastructure to various economic zones,
improving logistics efficiency

Kisan Rail and Krishi Udan: To assist farmers in
transporting perishables to distant markets
quickly.

Operation Greens: Focused on agri-logistics,
processing, and professional management for
tomato, onion, and potato (TOP) crops to
prevent distress sales

Outsourcing and PPP: Following the Shanta
Kumar Committee recommendations, the
stance is to outsource stocking and movement
operations to private players and state
agencies to bring down costs

way ahead

Containerization: Gradually shifting to
containerized movement of food grains to
reduce transit losses and improve handling
efficiency

Waterways: Exploiting riverine and coastal
waterways for cheaper and hassle-free
transport, especially for the North-East and
coastal states

Bulk Handling Integration:

Silos with Sidings: Ensuring that all new large
storage facilities (Silos) have railway sidings to
allow direct bulk loading and unloading,
eliminating multiple handling stages

Specialized Trucks: Operationalizing the use of
specially designed trucks for transporting grain
in bulk from mandis to silos

Digital Optimization:

Linear Programming: Utilizing computer-based
Linear Programming (LP) models to determine
the least-cost routes for movement. Currently,
compliance with LP plans is low (around 53%),
and improving this can save substantial freight
costs

Real-time Tracking: Implementing end-to-end
computerization and tracking of carriage to
minimize diversion and leakage

Village-Level Aggregation: Developing GrAMs
as aggregation hubs with transport links to
connect small farmers to the wider market
network

Cold Chain Expansion: Massive investment in
reefer vehicles and cold storage integration to
support the "farm to fork" cold chain,
particularly for the dairy, fishery, and
horticulture sector

CAG

storage of agricultural produce and issues and
related constraints

Status

The storage ecosystem comprises the Food
Corporation of India (FCI), Central Warehousing
Corporation (CWC), State Warehousing
Corporations (SWCs), and the private sector.

Types of Storage: The system primarily uses
Covered Godowns (bagged storage), Silos
(bulk storage), and Cover and Plinth (CAP)
(temporary open storage on raised platforms)

Utilization: There is a disparity in utilization. In
2021-22, FCI's owned capacity utilization was
low (56%) compared to hired capacity (86%)

Issues

Dependence on CAP (Open Storage):

Despite plans to phase it out, CAP storage is
still widely used, particularly in procuring
states like Punjab and Haryana.

for instance, 78% of damaged wheat stock in
Punjab was due to CAP storage

CAG

Regional Imbalances

There is a mismatch between procurement and
storage capacity. In Punjab and Haryana, FCI-
owned capacity is minimal compared to
procurement levels, leading to reliance on hired
godowns and CAP. Conversely, the North-
Eastern region faces an acute shortage of
storage capacity

Technological Obsolescence: The system
largely relies on manual handling ("human
back") rather than mechanized bulk handling
("machine back").

Old concrete silos in places like Gaya and
Kanpur have lain unutilized for 20 to 35 years

Liquidation Delays: Excess stocks over buffer
norms are not liquidated proactively, leading to
high carrying costs and storage crunches

Infrastructure Gaps: Many godowns lack
railway sidings, necessitating double handling
(loading/unloading at railheads and
transporting to godowns), which increases
costs

Post-Harvest Losses: Due to poor
infrastructure and handling, post-harvest
losses are estimated at around ₹50,000 crore
annually

NCF 4

Challenges

Land Acquisition: A major hurdle for
augmenting capacity (both conventional and
silos) is the non-availability of suitable land
and delays in acquisition.

Cold Chain Deficit: There is a massive shortfall
in cold chain infrastructure for perishables.
One study noted a 99.6% shortfall in integrated
pack-houses and an 85% shortfall in reefer
transport

Coordination Failures: There are instances
where silos are constructed but remain
unutilized because they lack necessary
dispatch facilities (like railway sidings) or
because bulk handling transportation is
unavailable

Economic Viability: Small farmers lack the
volume to utilize modern storage effectively on
their own. The distance to regulated markets
and lack of village-level aggregation centers
force distress sales

Govt stance

National Policy (2000): The government aims
to modernize handling and storage by
encouraging private sector participation
through schemes like Private Entrepreneurs
Guarantee (PEG) and developing bulk handling
silos

Decentralization: The policy encourages
Decentralized Procurement (DCP) where states
procure and store grains to reduce FCI's burden
and logistical costs

The government promotes Negotiable
Warehouse Receipts (NWRs) to allow farmers
to store produce in registered warehouses and
avail credit, preventing distress sales

Various funds like the Agri-Market
Infrastructure Fund (AMIF) (₹2,000 crore) and
Animal Husbandry Infrastructure Development
Fund (AHIDF) are instituted to create storage
and processing capacities

way ahead

Transitioning from gunny bags to bulk handling
across the supply chain

Aggressive expansion of steel silos (target of
100 LMT) on PPP mode to ensure better
preservation and reduced handling losses

Developing Gramin Agricultural Markets
(GrAMs) (22,000 targeted) as aggregation hubs
at the village level to feed into larger
warehousing and processing zones

Developing integrated cold chains (pack-
houses, reefer transport, ripening units) to
support high-value horticulture, meat, and dairy
sectors

e-tech in the aid of farmers e-technology in the aid of farmers

Examples

Precision Farming

Precision Farming (also called Satellite
Farming) refers to the application of a precise
and proper quantity of inputs like water,
fertiliser, pesticides, etc at the correct time to
the crop for increasing its productivity and
maximising its yields, by creating use of digital
farming technologies.

Tools

ICT, sensors (soil health, disease) , GPS ,
drones, variable rate tech, automatic
positioning systems

Benefits

reduced cost, improved yield, improved soil
health , optimised resource use (fertilisers,
water, electricity) , faster disease control

Digital Platforms & Connectivity:

The e-NAM (National Agriculture Market) has
integrated 1,409 APMCs, registering 1.75 crore
farmers and facilitating trade worth over ₹4
lakh crore

Kisan Sarathi, an online platform, connects
over 250 lakh farmers to experts via mobile
apps and IVR (Interactive Voice Response)
systems in 13 languages

The "Agri-Drone Project" is being implemented
to demonstrate drone usage for spraying
pesticides and nutrients.

Over 15,000 demonstrations have been
conducted across 31 states

Pest Management:

: An AI-based smart pheromone trap using the
YOLO algorithm is being used for real-time
monitoring of pink bollworm in cotton

Disease Identification: Mobile apps like AI-
DISC (identifies 70 crop diseases and 37 pests)
and AI-DISA (detects livestock diseases like
Mastitis and Lumpy Skin Disease) are in use

For livestock, the Bharat Pashudhan database
tracks over 75 crore animals.

Fisheries & Livestock: The National Fisheries

Digital Platform (NFDP) was launched to create
a registry of fish workers and enterprises

Drone

Garuda Aerospace – Uses Droni agricultural
drones for pesticide and nutrient spraying;

Robot

TartanSense (India) – Brat robot uses robotics
and computer vision for weed detection and
precision spraying in sugarcane and cotton
fields.

AI

Cropln (India) – SmartFarm & SmartRisk
platforms use Al for crop health monitoring,
yield forecasting, and climate risk assessment
for crops like rice, cotton, and soybean.

Microsoft - Al Sowing App (India)

Data-Driven Decision Making in Agriculture

PMFBY + Remote Sensing (India) - Uses
satellite data and yield models for faster and
more accurate crop insurance claims.

Challenges

Small Holding Size: With over 80% of farmers
being small and marginal, individual ownership
of high-tech machinery is economically
unviable. This necessitates shared economy
models like "Uberisation" of farm services

Lab-to-Land Gap: There is a wide gap between
scientific discovery and field adoption due to
lack of soft skills, insufficient demonstration,
and affordability issues

Infrastructure & Connectivity: Digital divides
and lack of physical infrastructure (like cold
chains and stable electricity) hinder the full
deployment of digital tools

Govt

The government aims to shift agriculture from
a "production-centric" to an "income-centric"
enterprise, leveraging technology as a key
enabler:

The government is building a "Digital
Agriculture Mission" (AgriStack) to provide
digital identities to farmers and integrate data
across schemes

Farming as a Service (FaaS): Promoting the
concept of FaaS to offer services like soil
testing, assaying, and machinery rental at
single-stop centers,

Blockchain & Traceability: Piloting blockchain
technology (e.g., for apples in Himachal
Pradesh) to ensure quality and traceability for
exports

Way ahead

Uberisation of Services: Aggregation platforms
for farm machinery and services should be
promoted to make technology accessible to
small farmers

Precision Agriculture: Widespread adoption of
sensors, AI, and drones for precise application
of inputs (water, fertilizer, pesticides) to reduce
costs and environmental impact

Integrated Digital Platforms: Seamless
integration of portals like e-NAM with logistics
and warehousing platforms to reduce market
information asymmetry

Adopting "Fork to Farm": Using digital tools to
reverse the approach, where market demand
dictates production decisions

Capacity Building: Large-scale e-literacy
campaigns to train farmers and extension
functionaries in using digital tools

Market-Led Extension: Extension services must
pivot to providing business-enabling
information (prices, logistics, buyers) using ICT
tools

Food processing and related industries in
India- scope’ and significance, location,
upstream and downstream requirements,
supply chain management.

Issues related to direct and indirect farm
subsidies and minimum support prices; Public
Distribution System- objectives, functioning,
limitations, revamping; issues of buffer stocks
and food security; Technology missions;
economics of animal-rearing.

Issues related to direct and indirect farm
subsidies and minimum support prices

Subsidy

Data

Total subsidy expenditure (FY 2025–26): ₹4.26
lakh crore

Increased from ₹4.03 lakh crore (FY 2023–24)

Food subsidy 2.03 lakh crore

Fertiliser subisdy

1.68 lakh crore

35% dependency met through imports

$9 Bn imports

Power Subsidy & Agriculture

Direct

Subsidies directly transferred to farmers,
usually in cash form with Aim to reduce
leakages and improve targeting

examples

PM-KISAN DBT of 6000 to farmers per year

Rythu Bandhu Scheme

India's first direct cash transfer scheme
providing financial support to farmers for farm
inputs like seeds, fertilizers, and labor, giving
them ₹5,000 per acre per season (Kharif &
Rabi), aiming to break debt cycles and boost
productivity without being a loan.

PAHAL (LPG DBT)

an Indian government initiative for the Direct
Benefit Transfer (DBTL) of subsidies for
Liquefied Petroleum Gas (LPG). The primary
goal is to ensure transparency and efficiency
by transferring the subsidy amount directly into
the bank accounts of eligible consumers.

Farm loan waivers

Benefits

Enhances purchasing power of farmers

Provides freedom of choice in expenditure

Improves resource-use efficiency

Farmers buy inputs (e.g., fertilisers) at market
price → avoid overuse

Cash transfers act as working capital for
agriculture

Lowers government administrative costs

Reduced storage, transport, and procurement
burden

Challenges to direct farm subsidies

Inadequate financial inclusion of farmers

Potential inflationary impact

Errors in identification of beneficiaries

Limited access to markets restricts effective
utilisation

Indirect farm subsidies

Subsidies where input prices are kept below
market prices

Provided through price controls or concessions

Types

Tax concessions

Insurance support MSP

Low-interest credit KCC credit cards

Depreciation write-offs

Rent and input rebates

Data

India spends around 2% of GDP on indirect
subsidies

examples electricity subsidy, fertiliser subsidy

MSP

Intro

MSP is the minimum price that the government
considers remunerative for agricultural
production. In event of fall of prices in notified
agricultural commodities, the government
provides price support by procuring the farm
produce at MSP.

Data

Covered 24

torai rapeseed and mustard

Dehusked Coconut Cpora

Post harvest losses 15-25% ICAR

commodities 23

Fair and Renumerative Price for sugarcane

recommended by Commssion for agri cost and prices (CACP)

fixed by Cbinet Committee on economic affairs

Benefits

ensures food secutity and affordability

Public Distribution System (PDS), supported by
procurement at MSP, has been central to
managing food economy

contain the rise in food prices and supply

It incentivizes cultivators to adopt modern
technology and raise productivity in line with
emerging demand patterns

during green revolution productivity enhanced
due to combination of MSP and tech
advancements

Indirect subsidies help stabilize markets for
basic foodgrains.

MSP protect farmers from sharp falls in prices,
particularly when international prices drop or
market volatility occurs

Protection from fluctuation in global markets
and food security is achieved

Alleviation of Rural poverty prevents rural to
urban migration for jobs

Challenges

Limited Reach and Inequity

Data from the 70th round of NSSO (2012-13)
revealed that only 6% of total farmers gained
from selling wheat and paddy directly to any
procurement agency

Regional Disparity:

Procurement operations are highly
concentrated in states like Punjab, Haryana,
Andhra Pradesh, and Madhya Pradesh

Fiscal Burden

High Cost: Food and fertilizer subsidies
constitute 87% of the total subsidy bill

4L cr

subsidies can distort market to ultimately hurt
the poor

Farm loan waiver not only increases the states’
fiscal deficit and interest burden, but also
limits their ability to invest in productive capital
in agriculture.

Environmental and Resource Inefficiency

Imbalanced Fertilizer Use:

The heavy subsidy on urea compared to
Phosphatic and Potassic fertilizers has created
a highly imbalanced use of Nitrogen,
Phosphorus, and Potassium (NPK)

Subsidies on chemical fertilizers make them
cheaper than organic options, creating an
uneven playing field

This discourages the use of organic manures
and affects soil health

Water Depletion:

Policies such as free power for agriculture,
combined with the dominance of water-

guzzling crops like paddy (incentivized by
MSP), have caused indiscriminate use of
groundwater.

This has turned large tracts of the Indo-
Gangetic Plains into "dark zones" due to over-
exploitation

Distortion of Cropping Patterns

skewed incentive structure where farmers
prefer wheat and rice crops over pulses and
oilseeds, even though the country is deficit in
the latter

Adverse effect on sugar mills

when State administrtive prices in set over and
above FRP

Operational Inefficiencies

Buffer stocks with the Food Corporation of
India (FCI) have often been more than double
the norms costing the nation thousands of
crores without serving a worthwhile purpose.

Storage Losses

Due to inadequate covered storage capacity,
substantial quantities of grains are stored in
Cover and Plinth (CAP) storage, leading to
deterioration

For instance, in Punjab, 78% of the damaged
wheat stock was attributed to CAP storage

Reforms

Minimum Support Price (MSP) and
Procurement

The Swaminathan Committee recommended
that MSP should be at least 50% more than the
weighted average cost of production

Diversification of Procurement:

government must provide better price support
for pulses and oilseeds and

dovetail MSP policy with trade policy to
prevent landed costs from falling below MSP

Decentralization:

handing over procurement operations of wheat
and rice to states with sufficient experience
(pujab and haryna)

shifting the Food Corporation of India’s (FCI)
focus to eastern states (Bihar, UP, West Bengal)
where farmers suffer from distress sales

Private Sector Participation:

The DFI Committee recommended permitting
private stockists to procure at MSP with valid
incentives

direct cash subsidy of about ₹7,000/hectare to
farmers

Shanta Kumar

Reforms

Agricultural Marketing and Logistics Reforms

The DFI Committee recommended a new
market architecture comprising 22,000 Gramin
Agricultural Markets (GrAMs) at the village
level (upgrading existing haats), linked to
competitive wholesale markets and export
markets

Single Indian Market

State APMC Acts should be amended to
promote grading, branding, and the
development of domestic and international
markets, moving towards a Single Indian
Market

Institutional and Structural Reforms

Restructuring FCI

FCI be restructured into an agency for
innovations in food grain management,

handle bulk supply chains

end to end computerisation

Land Reforms

It also suggested establishing a National Land
Use Advisory Service

distributing ceiling-surplus and waste lands
and preventing the diversion of prime
agricultural land for non-agricultural corporate
purposes

States should adopt the Model Contract
Farming and Services Act to facilitate high-
tech cultivation and better market linkages

Contract Farming: States should adopt the
Model Contract Farming and Services Act to
facilitate high-tech cultivation and better
market linkages

agriculture shall be placed in concurrent list of
constitutions

NITI Ayog recommendations Sustainability and Organic Inputs (Gaushalas)

Parity in Assistance: There should be parity in
support for chemical fertilizers vis-à-vis
organic/cow-dung-based fertilizers to create a
level playing field

Public sector fertilizer agencies (like IFFCO,
KRIBHCO) should be mandated to market
standardized organic and bio-fertilizers
produced by Gaushalas

The Fertilizer Control Order (FCO) should be
expanded to include products like
Panchagavya, Jivamrit, and Ghanjivamrut to
facilitate their commercial production and
quality control

Promote the production of Compressed Bio
Gas (CBG) using agri-biomass and cattle
waste, similar to the GOBARdhan scheme

Crowding Out Private Sector:

High statutory levies and commissions (Mandi
Tax) imposed by some states (e.g., up to 14.5%
in Punjab) on procurement crowd out the
private sector

bonuses declared by states over the MSP
distort the market, further discouraging private
participation

WTO Issues Related to India’s Farm Subsidies

Amber Box: Trade-distorting subsidies (e.g., MSP)

MSP above Deminimis level (10% of agri output
value)

Stockholding and MSP

Procurement at MSP alleged to distort global
prices

Green Box:

Minimally trade-distorting (e.g., R&D,
infrastructure)

DBT (Ryathu bandhu scheme)

Other issues

Failure of states to ensure rational use of
subsidised water and electricity

Public Distribution
System- objectives, functioning, limitations,
revamping;

PDS

Intro

Data

Beneficiaries

80Cr +

57% of pupulation

items rice, wheat, sugar, kerosene

NFSA

covering up to 75% of the rural and 50% of the
urban population

It categorizes beneficiaries into Antyodaya
Anna Yojana (AAY) (poorest of the poor) and
Priority Households (PHH)

Under the Pradhan Mantri Garib Kalyan Anna
Yojana (PMGKAY), the government provides 5
kg of free food grains per person per month to
priority households and 35 kg per family per
month to AAY households. This scheme was
extended to provide free grains effective from
January 2023

Issues

leakage

69000 Cr+

ICRIER study

28% of food grains

40-50% shanta kumar committe

Nutritional deficiency due to major empahsis
on rice and wheat

Fiscal impact ~2L Cr

tageting errors old data of 2011

storage losses 10% of annual spoilage

corruption at FRP underweighing, overchargins

govt stance

Tech reforms

Digitization of ration cards and beneficiary
databases to remove duplicates.

Mandatory Aadhaar linkage to prevent ghost
beneficiaries.

95% of Fair Price Shops are equipped with PoS
machines for biometric authentication

to reduce the leakage from

46% 2011-12

28% 2022-23 HCES, 2022-23

Ration Card Portability: One Nation One Ration
Card (ONORC): Enables access to PDS benefits
nationwide, aiding migrant workers.

Doorstep delivery of foodgrains to FPSs in
many states.

Nutritional Initiatives: Distribution of fortified
rice under POSHAN Abhiyan to combat anemia
and malnutrition.

Infrastructure Improvement: Construction of
modern silos, cold storage, and warehouses

Reforms

Reduce coverage from 67% to 40%, focusing on
BPL families.

Recent policy briefs suggest providing free
food only to the bottom 15% (extreme poverty),
while offering grains at 50% of MSP to the rest
of the beneficiaries

Increase grain allocation from 5kg to 7kg per
person

Shanta kumar committee

Fiscal Reforms Direct Cash Transfers:

Reforms suggest a gradual shift towards Direct
Benefit Transfer (DBT) or cash transfers for
food security, starting with large cities and
surplus states

curbing leakage, reduce administrative cost,
empower beneficiary to purchase grains of
their choice

Fair Price Shops (FPS) could be transformed
into nutrition hubs, potentially linked with a
food coupon system to ensure access to
diverse and nutritious food

A transparent liquidation policy is needed for
buffer stocks to prevent holding excessive
stocks that cost the nation heavily without
serving a purpose

Provide 6 months’ ration to beneficiaries after
procurement season. Reduce storage costs
and consumer hassles.

storage. Containerize grain movement to
minimize losses.

issues of buffer stocks and food security;

Buffer stocks

Data

furtis and veggies transport via roadways 97%

Requirement 70 mT

Availability 35 mT

Shanta Kumar Committe Report

Issues

Excessive Stockholding: A major challenge is
that buffer stocks held by the Food Corporation
of India (FCI) have often been more than
double the prescribed norms.

lack of norms for maxmimum level of stocks to
be maintained

Storage Capacity Constraints:

FCI's owned covered capacity is only 32% of
the total requirement

Consequently, substantial quantities of grain
are stored in Cover and Plinth (CAP) (open
storage) or hired godowns. In Punjab alone,
nearly 78% of damaged wheat stock was
attributed to CAP storage

Due to inadequate scientific storage and
prolonged storage of grain, significant
quantities are damaged

Between 2017-18 and 2021-22, 13,349 MT of
food grains valued at ₹35.57 crore were
damaged

CAG

Govt steps

Private Entrepreneur Guarantee (PEG) Scheme

private parties construct godowns which FCI
hires with a guarantee of usage

As of March 2022, 14.5 MMT capacity was
completed under this scheme

Silo Construction:

action plan to construct 100 LMT of steel silos
in Public-Private Partnership (PPP) mode.

progress has been slow

with only 12.25 LMT constructed by FCI
against a target of 29 LMT under the policy

Open Market Sale Scheme (OMSS)

government utilizes OMSS to offload excess
stock in the domestic market to control prices
and liquidate surplus

Price Stabilization Fund: A Price Stabilization

Fund is in place (with a budget of ₹4,019 crore
for 2025-26) to manage price volatility

Reforms required

A. Structural and Policy Reforms

Transparent Liquidation Policy: The Shanta
Kumar Committee (HLC) and CAG recommend
a pro-active liquidation policy that
automatically kicks in when stocks exceed
buffer norms. This involves selling surplus in
the open market (OMSS) or export markets to
reduce costs

Revisiting Buffer Norms: The norms need to be
revised to clearly delineate between
operational stocks and strategic reserves.

A maximum stock limit should be prescribed
to prevent unlimited accumulation

Decentralized Procurement: The HLC
recommended handing over procurement of
wheat and rice to states that have gained
sufficient experience (e.g., Punjab, Haryana,
MP) and focusing FCI’s attention on deficit
states to prevent distress sales

B. Infrastructure and Storage Reforms

Phase Out CAP Storage: Covered and Plinth
(CAP) storage should be gradually phased out,
with no grain remaining in CAP for more than 3
months. It should be replaced by silos and
conventional godowns.

• Shift to Silos and Bulk Handling: There should
be a massive shift from bagged storage to bulk
handling using steel silos. This promotes a
"machine back" rather than "human back"
approach, reduces losses, and improves shelf
life.

• Private Sector Participation: FCI should
outsource stocking operations to the Central
Warehousing Corporation (CWC), State
Warehousing Corporations (SWC), and the
private sector on a competitive bidding basis to
reduce costs

Agri-Logistics: The DFI Committee
recommended promoting integrated agri-
logistics (pack-houses, reefer transport) to
reduce post-harvest losses and facilitate the
"Fork to Farm" approach

C. Market Reofrms

Negotiable Warehouse Receipts (NWRs): The
NWR system should be scaled up. Farmers can
deposit produce in registered warehouses and
get bank advances (say 80% of MSP) against it.
This reduces the need for physical
procurement by the government and allows
farmers to sell when prices are better.

• Direct Benefit Transfer (DBT): A gradual shift
to cash transfers in the Public Distribution

System (PDS) is recommended for large cities
and surplus states. This would reduce the
physical handling of grain, plug leakages
(saving an estimated ₹30,000 crore annually),
and reduce the pressure on buffer stocks.

• Movement Optimization: FCI should strictly
follow Linear Programming to minimize the
cost of transporting grains. Currently,
deviations from optimal routes result in higher
freight costs. Movement should be gradually
containerized to reduce transit losses

Food Security

Definition

when all people at all times

have physical and economic access to
sufficient, safe and nutritious food to meet
their dietary needs and food preferences for an
active and healthy life.

Intro

While India has achieved self-sufficiency in
grain production, food security challenges have
shifted from availability to accessibility,
nutritional quality, and system efficiency.

Issues

Calorie vs. Nutrition: The current system is
heavily cereal-centric (rice and wheat), which
provides calories but lacks essential
micronutrients. This has led to "hidden hunger"
or micronutrient deficiency

35.5% of children under five are stunted, 19.3%
are wasted, and 32.1% are underweight

Dietary Imbalance: Household consumption
surveys (2022-23) reveal that the share of
spending on nutritious items like pulses and
vegetables in total food expenditure has
declined, while spending on processed foods
has increased

leakage 28% of PDS 69000 cr

regional disparity

Leakages are significantly higher in certain
states, such as Arunachal Pradesh (63.2%) and
Nagaland (60.4%), compared to states with
better tracking

Coverage Errors: There are issues of both
exclusion (needy excluded) and inclusion (non-
eligible included) errors in the beneficiary lists

Govt steps

govt steps under PDS sections

nfsa, pmgkay,digitisation, epos, one nation one
ration card

Poshan Abhiyan: Initiatives like Saksham
Anganwadi and POSHAN 2.0 have been
allocated ₹21,960 crore (2025-26 BE) to
improve nutritional outcomes for children and
mothers

Bio-fortification: The government is promoting
bio-fortified varieties of crops (e.g., zinc-rich
rice, iron-rich pearl millet) to address
micronutrient deficiencies

reforms

PDS related

dbt , reduction in coverage, updated beneficiary
list, diversification , food coupan and stamps at
FPS

storage reforms

silo, containsersiation, warehouse, farm to fork
infra, etc.

shift towards millets Intl. year of millets 2018

Life-Cycle Approach: Reorganizing nutrition
support programmes on a life-cycle basis,
targeting specific nutritional needs from
infancy to old age

Community Food Banks: Promoting the
establishment of Community Food and Water
Banks operated by Women Self-help Groups
(SHGs) to ensure local food security

National Food Guarantee Act: The
Swaminathan Committee suggested
formulating a National Food Guarantee Act that
combines the features of food for work and
employment guarantee programmes

FAO

economics of animal-rearing.

Livestock

Status

This accounts for 30.23% of the Agricultural &
Allied Sector GVA and 5.50% of the total
national GVA,. The sector grew at a Compound
Annual Growth Rate (CAGR) of 7.67% between
2014-15 and 2021-22 at constant prices

Milk

Milk: India is the largest milk producer in the
world, contributing 25% to global production

Eggs: India ranks 2nd globally in egg production

India ranks 4th globally in meat production,

The sector provides livelihoods to
approximately 8.5 crore rural households

DAHD annual report 2024-25

Importance

boost exports positively impacting BoP leather, meat, wood, etc.

empolyement generation in rural households

integerated farming would lead to reduction of
chemical fertilisers reducing import bill and
improving soil health

combating hidden hunger

Risk mitigation by providing a sense of security
and confidence to landless, small, and
marginal farmers

Biogas production and sustainable agricultural
development

Issues

Feed and Fodder Deficit: There is a critical
shortage of feed. The country faces a net
shortfall of 35.6% in green fodder,

Low Productivity: Despite high total production,
per-animal productivity remains a challenge.
For instance, the average milk yield for
indigenous cows is significantly lower (4.20
kg/day) compared to exotic cows (9.82 kg/day)

Disease Outbreaks: Diseases like Foot and

Mouth Disease (FMD) and Brucellosis cause
massive economic losses, estimated at around
₹24,000 crore annually for FMD alone

Stray Cattle and Unproductive Animals:

male cattle and unproductive cows are often
abandoned, leading to a stray cattle population
estimated at 53 lakhs,. This causes crop
damage and road accidents

Artificial Insemination (AI) coverage is limited
to 33% of breedable bovines, with the
remaining 70% covered by scrub bulls of
unknown genetic merit

Viability of Gaushalas: Most Gaushalas (cow
shelters) depend on donations and lack
sustainable revenue models. Operating
expenses often exceed income, making them
financially vulnerable

DAHD

DAHD

NITI aayog

Govt stps

Rashtriya Gokul Mission (RGM): Launched to
develop and conserve indigenous bovine
breeds. It focuses on enhancing productivity
through technologies like IVF and Sex Sorted
Seme

susbidies for establishment of Breed
Multiplication Farms and for IVF pregnancies

National Livestock Mission (NLM)

Realigned to focus on entrepreneurship
development in poultry, sheep, goat, and
piggery

It provides capital subsidies (up to 50%) for
establishing breeding farms to individuals,
FPOs, and SHGs

Health

National Animal Disease Control Programme
(NADCP): A central sector scheme to control
FMD and Brucellosis through mass vaccination
(e.g., FMD vaccination by 2025 and eradication
by 2030)

Pashu sanjeevni

animal welness programme encompassing
provision of animal health card along with UID
identifications

National Aritificail Insemination Programme

to prevent spread of disease and sex-sorted
breeding

Animal Husbandry Infrastructure Development
Fund (AHIDF)

to incentivize investments in dairy/meat
processing, animal feed plants, and waste-to-
wealth management. It offers 3% interest
subvention on loans

Kisan Credit Card (KCC): The facility of KCC
has been extended to animal husbandry and
fisheries farmers to meet their working capital
needs

Digital Initiatives: Launch of the "Bharat
Pashudhan" database and the 1962 mobile app
to create a digital ecosystem for livestock
management

Mobile Veterinary Units (MVUs): Financial
support is provided to states for MVUs to
deliver veterinary services at farmers'
doorsteps

E pashu haat protal

connecting breeders and farmers regarding
availability of bovine germplasm

reforms required

Promotion of Sex Sorted Semen: To address
the issue of stray cattle and male calves, the
widespread use of sex-sorted semen
(producing female calves with >90% accuracy)
is recommended.

"Cow Economy" and Waste to Wealth:

Bio-Fertilizers: Promoting the production of
Phosphate Rich Organic Manure (PROM) and
vermicompost from cow dung to make
Gaushalas economically viable and reduce
reliance on chemical fertilizers

Bio-Energy: Encouraging the setting up of Bio-
CNG and biogas plants using cattle waste,
linked with schemes like GOBARdhan and
SATAT

Fodder Development: utilizing non-arable and
wastelands for fodder production through
agroforestry to bridge the supply deficit.
Incentivizing fodder storage (silage/hay)
entrepreneurs

Vetenary

One Health Approach: Strengthening the "One
Health" framework to mitigate risks of zoonotic
diseases (diseases transmitting from animals
to humans) through integrated surveillance and
"Standard Veterinary Treatment Guidelines"

Institutional Strengthening: Forming Village
Producer Organisations (VPOs) for small
ruminants and poultry to improve access to
credit and markets,. Strengthening the
cooperative credit structure for long-term
lending to the sector

Export Focus: Improving compliance with food
safety standards of Codex Alimentarius
Commission to boost exports of dairy, meat,
and poultry products, where India currently has
untapped potential

NITI Ayog

bredding indegeneous cattles with exotic to
enhance producitivity

Aquaculture

Status

second-largest aquaculture producer and the
4th largest exporter of fish and fisheries

Structural Shift: There has been a significant
shift from marine capture to inland
aquaculture. Inland fisheries now constitute
about 75% of total production, a major increase
from 50% in 2000-01

7.26% of the agricultural GVA

annual average growth rate of 6.30% over the
last five years

Exports USD 7.38 billion FY 2023-24

The sector supports the livelihood of
approximately 28 to 30 million people

Issues

Infrastructure Deficits:

insufficient cold chain connectivity from
landing sites to retail markets, leading to post-
harvest losses and hindering domestic demand
in the hinterlands

The lack of strong value chains also hampers
export competitiveness

lack of specifically fishing harbours, landing
centers, cold chains, and distribution systems.

leads to post-harvest losses, poor processing,
and value addition

Technological Gaps:

shortage of quality fish seeds, lack of mapping
for fish-rich zones, and the use of obsolete
fishing technology (bottom thrwaling)

Ecological Concerns:

Over-exploitation of marine resources and
depletion of inland natural waters are major
concerns

Pollution: Marine plastic litter is a growing
threat

Marine capture fisheries face limited scope for
expansion due to overcapacity in territorial
waters, weak regulation, and prevalence of
traditional fishing practices

Marketing and Financial Issues:

The share of high-value processed products is
low (<15%) compared to global competitors
like China (49%)

Inland and Aquaculture Challenges:

Resource Issues: Seasonal nature of
operations, depleted stocks in natural waters

Input Constraints: Shortage of quality seed
and feed, low productivity, and high input costs

Freshwater aquaculture is heavily dependent
on Indian Major Carps (Catla, Rohu, Mrigal),
lacking diversification

Export Quality Concerns:

"Salmonella" and veterinary drug residues are
primary causes for export rejections in
crustaceans

Sustainability Threats: The sector faces
threats from marine plastic litter, pollution, and
climate change affecting ecosystems

Govt steps

Pradhan Mantri Matsya Sampada Yojana
(PMMSY):

address critical gaps in the value chain,

Infrastructure Development (FIDF): The
Fisheries and Aquaculture Infrastructure
Development Fund (FIDF) (fund size ~₹7,522
crore) provides concessional finance to state
governments and private entities for
developing fishing harbours, cold storages, and
feed mills

Kisan Credit Card (KCC): Extended to fisheries
farmers to meet working capital needs. The
lending limit was recently increased to ₹5 lakh
in the 2025-26 budget

Technology and Modernization:

Launch of the National Fisheries Digital
Platform (NFDP) to create a registry of fish
workers and enterprises

Vessel Communication: National Rollout Plan
for Vessel Communication and Support
System (VCSS) involving transponders for
safety and communication

Disease Monitoring: Launch of the Report Fish
Disease (RFD) App and the National
Surveillance Programme for Aquatic Animal
Diseases (NSPAAD)

Disease Monitoring: Launch of the Report Fish
Disease (RFD) App and the National
Surveillance Programme for Aquatic Animal
Diseases (NSPAAD)

Implementation of the National Plan of Action
on Sharks and engagement in the GloLitter
partnership to reduce marine plastic

MGNREGS for developing fish ponds

Reforms required

Legal and Regulatory Framework:

Enactment of a comprehensive National
Marine Fisheries (Regulation and
Management) Bill to regulate the Exclusive
Economic Zone (EEZ)

Adoption of a Model Leasing Policy by states
to facilitate mariculture and secure tenure
rights for small-scale fishers

Shift to Blue Economy and Deep Sea Fishing:

Encouraging traditional fishers to undertake
deep-sea fishing for high-value resources like
Tuna (especially in Andaman & Nicobar and
Lakshadweep)

Designating areas for Mariculture Technology
Parks and promoting seaweed cultivation

Value Chain and Marketing:

Cluster Approach: Developing integrated
production and processing clusters to enable
focused development of export-oriented
species

Traceability: Implementing IT-enabled
traceability and labeling systems from 'bait to
plate' to ensure quality and meet international
standards

Cold Chain: Massive expansion of cold chain
infrastructure to reduce post-harvest losses
from the current high levels to below 10%

Institutional Strengthening:

Promotion of Fish Farmer Producer

Organizations (FFPOs) to enhance bargaining
power and market access

Establishment of a National Fisheries
Development Council for overall guidance and
coordination between Center and States

Diversification: Moving beyond carps to culture
exotic species and high-value native species
(like Scampi, Tilapia, and Trout) in inland waters

Technology missions; To be done with cropping , irrigation, etc.

Food processing and related industries in
India- scope’ and significance, location,
upstream and downstream requirements,
supply chain management.

Food Processing

Intro

We export what we should eat and waste what
we shoudl Process

The food processing sector is recognized as a
"sunrise industry" with immense potential for
value addition and employment generation

Conclusion

Backbone of AtmaNirbhar krishi, potential to
linking farms into international food chains and
reduce rural distress.

Data (Scope and significance)

GDP Contri

Manufacturing 7.7-8.8% Delloite and FICCI Report

5.5% of total GVA

CAGR (2015-16 - 2022-23)

5.35%

1.61 -> 1.92 L Cr.

MOSPI

% of total agri exports

23%

Rising CAGR @13%

13 Bn in 2022-23

% of total organised employement

12%

70L

Investment growth in last 7 year 7.25%

Production base

India is largest producer of milk, spices, pulses

India is 2nd largest producer of fruit ,
vegetable, fish and eggs

It is crucial for doubling farmers' income by
reducing post-harvest losses (estimated at
₹50,000 crore annually) and providing better
price realization

FDI 100% allowed

provides raw material base for processing

Location

The industry is concentrated in coastal states
due to accessibility to marine resources and
ports. Major hubs include Andhra Pradesh
(14.3% of registered factories), Tamil Nadu
(12.7%), Telangana, Maharashtra, and Gujarat

Specific Clusters: Bakery manufacturing is
prominent in Delhi, Uttar Pradesh, West Bengal,
and Maharashtra

Upstream and Downstream Requirements

Upstream (Farm to Factory):

Inputs: Accessibility to high-quality raw
materials (crops, meat, fish) and modern
extraction techniques

Linkages: Strong direct linkages with farmers
to ensure steady supply, bypassing
intermediaries (e.g., contract farming)

Infrastructure: Storage facilities for raw
materials, sorting/grading facilities, and
transport connectivity

Downstream (Factory to Consumer):

◦ Processing: Deployment of latest processing
techniques and machinery

Distribution: Organized retail networks, cold
chains for finished goods, and digital platforms
for faster distribution

Quality Control: Testing facilities to ensure
safety standards

Supply Chain Management (SCM)

Fragmented Nature: The chain is currently
characterized by multiple intermediaries
(middlemen), leading to inefficiencies, high
costs, and wastage

Logistics Costs: Logistics costs in India are
high at 13–14% of GDP compared to the global
average of 8–10%

Strategic Need: Efficient SCM is vital to
balance supply with demand, reduce
perishability, and improve profitability

Status

Dairy: Largest commodity, with India
contributing 25% of global milk production

Nutraceuticals: Valued at USD 8 billion,
expected to grow at 11% CAGR

Budget

National Institute of Food Technology,
Entrepreneurship and Management to be set
up in Bihar

Post Harvest Losses in Perishables 20%

Issues

Low Level of Processing

India 10% of agri produce

Developed countries 60-80%

Fragmented supply chains

lack of storage 37 mt

Cold Chain Deficit: A study indicated a 99.6%
shortfall in integrated pack-houses and 85%
shortfall in reefer transport vehicles

lack of processing infra

testing labs, effluent treatment plants, modern
processing units, etc.

Credit access 6% of MSME in food processing recive it

Lack of Skilled Manpower

80% unskilled/ informal

Skill Deficit: Severe shortage of skilled
manpower; NSDC estimated a need for 17.8
million skilled workers by 2022

NSDC , 2023

Red Tapism

Quality & Safety: Large unorganized sector
(75% of units) makes enforcing quality
standards (FSSAI/Codex) difficult, leading to
export rejections due to pesticide residues and
microbial contamination

Govt Stance

Economic

FDI 100% thtough automatic route

PLI

Outlay of ₹10,900 crore to create global food
manufacturing champions and support Indian
brands abroad

in food processing sector (Mozarella Cheese)

In

Ready to eat

Millet

PSL

Low GST 0/5%

Infrastructure Funds: Establishment of

dedicated funds like FIDF (Fisheries) and DIDF
(Dairy) to boost infrastructure

Operation Green

Focused on stabilizing the supply and prices
of Tomato, Onion, and Potato (TOP) crops and
expanded to 22 perishables

Originally for TOP

Now expanded to (TOTAL) 22 perishable items

Aim Price Stabilisation

method Integerated logistics development

support of FPO

modernise agri logistics

processing facilities

managagement

Pradhan Mantri Formalisation of Micro Food

Processing Enterprises Scheme (PMFME)
Scheme: Supports formalization of 2 lakh
micro-enterprises with credit-linked subsidies
and a "One District One Product" (ODOP)
approach

Credit Linked Subsidy

Common Infrastructure

ODOP

Seed Capital for SHGs

Seed capital @ Rs. 40,000/- per member of
SHG for working capital and purchase of small
tools maximum upto 4 Lakhs per SHG.

Marketing & Branding

The scheme provides 50% financial grant for
Branding and Marketing

Capacity Building

The Capacity Building component under
PMFME Scheme envisages to provide training
on Food Processing Entrepreneurship
Development Programme to the PMFME
beneficiaries.

PM-Kisan sampada Yojna

Cold storage, food parks, etc.

Investment 10,000 Cr

Jobs generated 6L

Market

ODOP

e-NAM

Agri Infra Fund

1L Crore

Farm gate Infra

Primary Processing

Food processing fund by NABARD worth 2000
crore

Need to be Done

Agri-export zone

Hub and Spoke Model: Developing a grid of
aggregation hubs (GrAMs) at the village level
linked to large processing centers

Cluster Developement JackFruit Cluster Kerala

Credit acess

Strengthen FPO and contract farming

Skill Development under Skill India Mission

Food Safety

Packaging

Tech Integration: Adoption of AI, IoT, and
Blockchain for traceability, demand
forecasting, and inventory management

Value Addition: Shifting focus from primary
processing to secondary and tertiary
processing (ready-to-eat, nutraceuticals) to
capture higher value

Public-Private Partnerships (PPP):
Encouraging PPP models for developing
storage, cold chains, and modern markets to
leverage private sector efficiency

Land reforms in India.

Land Reforms

Data

Landless Households [Socio Economic Caste Census 2011] 30%

Negative

%person own less than 2 hectare [NSSO] 90%

agri cencus 2018 farmer data

Landless 7

Marginal 75

Small 10

Positive Small and marginal owned area

1961 20%

2013 53%

In 1991-92, the bottom 50% of rural
households owned only 3% of the total land,
while the top 10% owned 54%

Fragmentation: The average size of operational
holdings has declined significantly, from 2.67
hectares in 1960-61 to 1.34 hectares in 1990-
91

In 1991-92, 11.24% of rural households were
strictly landless

Currently, small and marginal farmers own 86%
of the land in India

Model Contract Farming and Services Act, 2018

NCF

NCF

NCF

Introduction

The agrarian economy of India has historically
been characterized by inefficiency, low
productivity, and small units of production

Agrarian Reforms Committee (Kumarappa
Committee) recommended in 1949 that there
should be no place for intermediaries and that
"land must belong to the tiller". It viewed land
reforms as the "touchstone of democracy"
essential for freedom from poverty

Issues

The colonial land tenure systems—Zamindari,
Mahalwari, and Ryotwari—created a disconnect
between the cultivator and the land, leading to
exploitation

Reforms so far

Abolition of Intermediaries (1950-1972):
Removing Zamindars and Jagirdars to bring
cultivators in direct contact with the State

Intermediaries were abolished over 6.0 million
hectares, and ownership rights were granted to
20 million tillers

Tenancy Reforms: Providing security of tenure,
regulating rents (recommended at 20-25% of
gross produce), and conferring ownership
rights to tenants

Rights of 12.42 million tenants were protected
over an area of 6.25 million hectares

successful in kerala and WB

Operation Barga: In West Bengal Operation
Barga was launched in 1978 with the objective
of achieving the registration of sharecroppers
and providing them permanent occupancy and
heritable rights and a crop division of 1:3
between landowner and sharecropper.

Land Ceiling Acts: Fixing statutory limits on
land ownership and redistributing surplus land
to the landless

Redistribution: An area of 2.12 million
hectares of ceiling-surplus land was
distributed to 5.67 million rural poor (51% of
whom were SC/ST)

Consolidation of Holdings: Merging scattered
landholdings into compact blocks to improve
farming efficiency

About 65.34 million hectares of land was
consolidated, notably in Punjab, Haryana, and
Uttar Pradesh

Forest Rights Act, 2006: Recognized rights of
forest-dwelling Scheduled Tribes and other
traditional forest dwellers

Right to Fair Compensation and Transparency
in Land Acquisition, Rehabilitation and
Resettlement Act, 2013 (LARR Act):

Replaced the colonial 1894 Act to ensure
humane, participatory processes and fair
compensation for land acquisition

Recent steps

Model Agricultural Land Leasing Act, 2016:
Proposed by NITI Aayog to legalize land
leasing, ensuring security for land owners and
access to credit for tenants

Model Land Titling Act: The central
government has drafted a Model Land Titling
Act, aiming to standardise procedures and
facilitate a shift towards conclusive titling.

The Registration Bill 2025: Aimed at
streamlining land transfers and ensuring that
registration reflects actual ownership, not just
transactions

DILRMP (Digital India Land Recored
Modernisation Programme)

Digital India Land Records Modernization
Programme (DILRMP): Aimed at digitizing land
records

ULPIN / Bhu Aadhar 14 digit alpha-numeric for each land parcel

GPS Based

integerated with land records

As of recent data, 94% of Rights of Record
(RoR) have been computerized

2024, MoRD

PM SVAMITAVA

6.62 lakh villages in the country which will be
eventually covered in this scheme.

clear ownership of property in 'Abadi' in habited
area

It is a scheme for mapping the land parcels in
rural inhabited areas using drone technology
and Continuously Operating Reference Station
(CORS) to provide property cards.This Scheme
comes under the Ministry of Panchayati Raj.

MoRD

Factors Responsible

reforms succeeded (e.g., West Bengal's
Operation Barga, Kerala),

Strong political commitment allied with
peasant movements challenged the resistance
of landed interests

Record of Rights: Successful states prioritized
the updating of land records (e.g., recording
sharecroppers in West Bengal)

Decentralization: Involvement of Panchayati
Raj Institutions (PRIs) in identifying
beneficiaries and managing land

Kisan Sabhas and Farmers Associations also
helped farmers organize themselves and raise
their demands.

Judicial backing and progressive
interpretations of constitutional provisions
aided in land reforms. Egland reform
legislations through IX schedule.

Failure

Lack of Reliable Records: The absence of
accurate land records made it difficult to
identify surplus land or protected tenants

Legal Loopholes & Litigation: Large amounts of
surplus land remained locked in court cases.
Landowners used loopholes like Benami
transactions to evade ceiling laws

Administrative Apathy: The revenue
administration often lacked the vision and
dynamism required for a holistic view of land
management

Alienation of Tribal Land: There has been
widespread dispossession of Dalits and Tribals
from lands legally allotted to them

NCF

Land Reforms and Agriculture Productivity

Debate

higher farm szie enhances productivity due to
mechanisation, and economy of scale

small farm size increase productivity due to
resource efficiency, high labor intensity and
supervision

Productivity is linked to the size of holding;
Indian productivity is much lower than global
peers (e.g., paddy yield is 2929 kg/ha in India
vs. 6321 kg/ha in China)

Reforms

Security of tenure encourages investment in
land improvement. However, excessive
fragmentation (minuscule holdings) hampers
modern technology adoption To address this,
contract farming and producer organizations
(FPOs) are recommended to aggregate
produce without dispossessing farmers of their
land ownership

Land Records

Issues

Records are often outdated and do not reflect

ground realities (e.g., sharecroppers not
recorded).

60% of holdings still lack "conclusive titles,"
leading to endless litigation.

Discrepancy between revenue records and
registration data

Slow progress in the survey/resurvey of land
parcels

multiple claims

Complex inheritance and land transfer systems
often result in overlapping claims and
ownership ambiguity.

NCF

Land tilting

Current Status (Presumptive Titling)

India follows a system where land records are
fiscal (for revenue) and presumptive.

Registration of deeds under the Registration
Act (1908) records the transaction, not the title
itself.

Ownership is proven through a chain of
documents

Reforms Required (Conclusive Titling): There is
a move toward Conclusive Titling, where the
State guarantees the title. This requires a
single, integrated agency for maintaining land
records

Benefits

Reduced Litigation: More than 55% of civil
disputes in India are land-related. Clear titles
would drastically reduce this litigation

Access to Credit: Clear titles facilitate easier

access to institutional credit (mortgages)

Greater Market Efficiency: Accurate and
transparent land records can facilitate land
transactions and promote a more efficient land
market.

Challenges: The transition requires massive
investment in surveying, updating records, and
legal changes.

Current Issues

Restrictive Tenancy Laws: Bans on leasing in
many states prevent land from being used
efficiently and leave tenants without legal
protection

Kerala

Exclusion: Women, despite legal rights (Hindu
Succession Amendment Act 2005), often lack
actual control or titles (Joint Pattas)

women constitute only 14% of all operational
landholders

Financial Exclusion: Tenant farmers cannot

access formal credit (Kisan Credit Cards) due
to lack of land titles

way ahead

legal front

Legalize Land Leasing: States must adopt the
Model Agricultural Land Leasing Act, 2016 to
allow land owners to lease out land without
fear of losing ownership and to enable tenants
to access credit/insurance

Special Land Tribunals: Establish fast-track
tribunals to resolve the 7.4 lakh acres currently
locked in litigation (Economic Survey 2024-25
context).

Inclusive Definition of Farmer: Delink the
definition of "farmer" from land ownership to
include tenant farmers, sharecroppers, and
women for accessing government benefits

Innovative

Land Pooling: Promote Farmer Producer
Organizations (FPOs) and group farming to
create economic scales of operation for
smallholders

Contract Farming: Adopt the Model Contract
Farming Act to link farmers with markets and
technology while protecting land rights

Digitization: Complete digitization of land
records on a "Mission Mode" including geo-
tagging to reduce disputes

Digitizing Maps: Completing the 68%
digitization of cadastral maps (DoLR 2024) to
accurately identify encroached or surplus
government land.

Administrative changes at the state level that
streamline the collection and maintenance of
land data, for example - Karnataka’s BHOOMI
Project.

Draft National Land Reforms Policy: The draft
has five goals

Restore land unjustly taken from vulnerable
communities such as Dalits (untouchables)
and Tribals.

Protect the land of the Dalits and Tribals
including the commons that they depend on
going forward

Liberalize leasing laws.

Improve land rights of women.

Appropriate amendment to Forests Right Acts,
2006.

Implementation Good Implementation

Kerala (EMS Nambodiriprasha)

WB (CPIM)

Challenges in digitisation

Farmer Debt

% of household spend more than they earn 70%

% in debt due to rise in farm input cost 50%

NSSO data

Economic Survey 2024

· · ·

Technology

· · ·

Environment (Conservation,
environmental pollution and

degradation, environmental impact

assessment) ···

Diaster Management

· · ·

Security